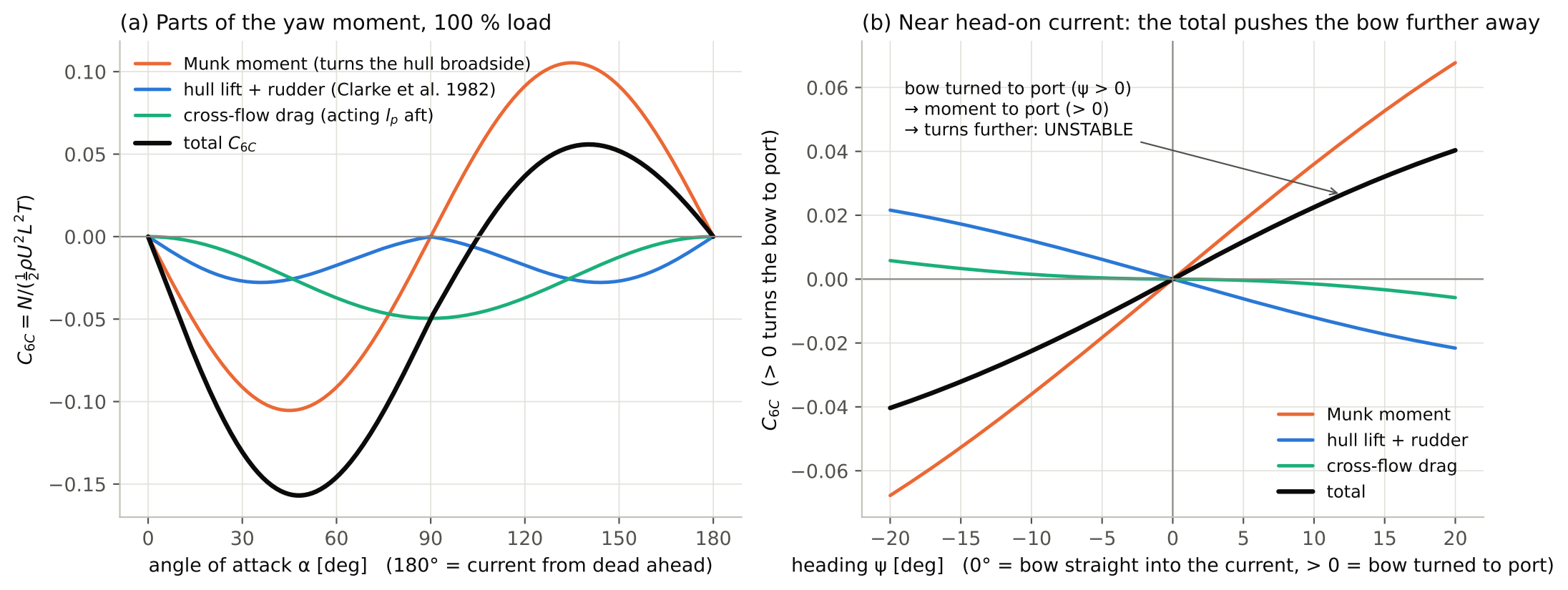
(a) Parts of the yaw moment, 100 % load
Munk moment (turns the hull broadside)
hull lift + rudder (Clarke et al. 1982)
cross-flow drag (acting l_p aft)
total C_{6C}
$C_{6C} = N/(\frac{1}{2}\rho U^2 L^2 T)$
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
30
60
90
120
150
180
angle of attack α [deg] (180° = current from dead ahead)
(b) Near head-on current: the total pushes the bow further away
bow turned to port (ψ > 0)
→ moment to port (> 0)
→ turns further: UNSTABLE
C_{6C} (> 0 turns the bow to port)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
Munk moment
hull lift + rudder
cross-flow drag
total
−20
−15
−10
−5
0
5
10
15
20
heading ψ [deg] (0° = bow straight into the current, > 0 = bow turned to port)